

SingleRAN

Security Overview Feature Parameter Description

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Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue does not introduce any changes to SRAN21.1 01 (2025-03-10).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

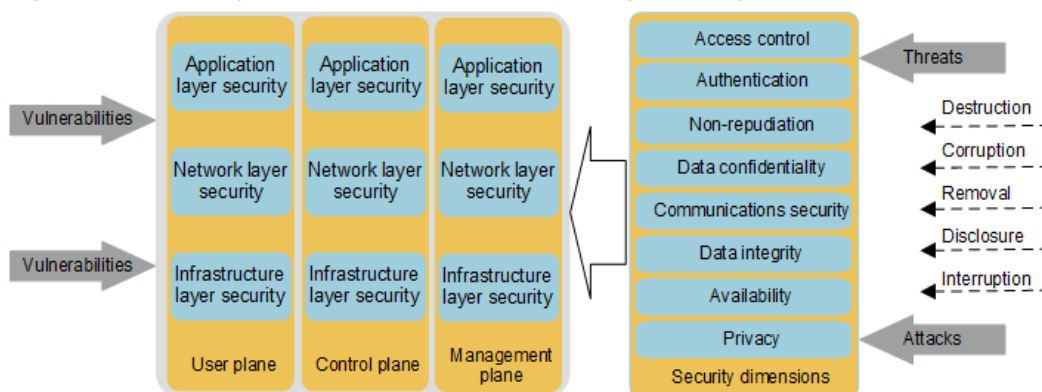
For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

3 Security System

The evolution to IP networks has increased network bandwidth and reduced operator's operating expense (OPEX). However, inherent vulnerabilities of IP networks, which are introduced to radio networks, leave them susceptible to security threats. In addition, more smart wideband mobile terminals pose new security challenges to radio access networks despite their diversified communications services and better user experience.

To address the increasing severity of network security threats, Huawei has designed and deployed SingleRAN networks to provide multi-plane and multi-layer security protection against various threats and attacks. This design is in accordance with the security model defined in ITU-T X.805. [Figure 3-1](#) shows the security architecture of a Huawei SingleRAN system.

Figure 3-1 Security architecture of a Huawei SingleRAN system



The security architecture of a Huawei SingleRAN system defines three security planes, three security layers, eight security protection dimensions, and five security threats. In addition, this security architecture introduces security management and security principles and policies as follows:

- Security plane: management plane, control plane, and user plane
- Security layers: infrastructure layer security, network layer security, and application layer security
- Security dimensions: access control, authentication, non-repudiation, data confidentiality, communications security, data integrity, availability, and privacy

- Security threats: destruction, corruption, removal, disclosure, and interruption
- Huawei security management applies to complex communications scenarios where network management security must be improved.
- Huawei security principles and policies are applied throughout the design and implementation of security solutions.

3.1 Security Threats

3.1.1 Common Security Threats

The ITU-T X.805 security architecture defines security threats as follows:

- **Destruction**
Destruction involves damage to physical equipment, information, and other resources. Communications equipment is generally located at a specific site or in an equipment room, which must provide required temperature and humidity for proper working of the equipment. Broken equipment room may result in communications equipment damage. The site where an equipment room is located must also be secure to prevent unauthorized access.
Door status sensors, temperature detectors, smoke detectors, and other detectors are specifically configured for an individual site or equipment room. These sensors are programmed to alert OM personnel in case of any onsite exceptions.
- **Corruption**
Corruption mainly involves tampering with information assets. Communications networks are managed by OM systems. Using accounts and passwords, authorized OM personnel log in to OM systems to set, modify, and remove data. Hackers attempt to decrypt accounts and passwords by brute force attack and modify key information, such as system configurations, which undermines normal system running.
For this reason, role-based rights management is implemented and account and password policies are correctly configured to protect accounts and passwords of OM systems.
- **Removal**
Removal includes theft, elimination, or loss of information or other resources. Hackers attempt to decrypt accounts and passwords and perform other malicious operations including stealing or clearing key information, such as system configurations, which undermines normal system running.
For this reason, role-based rights management is implemented and account and password policies are correctly configured to protect accounts and passwords of OM systems.
- **Disclosure of information**
Disclosure of information mainly refers to disclosure of information that is transmitted on networks, especially on an IP network, and eavesdropping on the radio interface. A hacker uses a network sniffer to intercept and analyze data flows in search of valuable information such as user names, passwords, or even conversations. An effective method for preventing information disclosure during data transmission on networks is data encryption, such as

adopting Secure Sockets Layer (SSL) and Internet Protocol Security (IPsec). With such a method used, hackers cannot obtain plaintext information even if interception occurs on networks.

Hackers also attempt to intercept communications over the radio interface. If user data is not encrypted or is weakly encrypted, hackers may decrypt communications data using standard protocol-defined data formats and algorithms. To prevent illegal eavesdropping, radio interface encryption standards are formulated and strong radio interface encryption algorithms can be used to secure user communications data transmitted over the radio interface.

- Interruption of services

By exploiting system vulnerabilities, hackers can launch attacks on the network to interrupt radio services. Network elements (NEs) on an IP-based communications network (especially a bearer network using leased lines) are vulnerable to cyber attacks. These attacks include denial of service (DoS), malformed packets, User Datagram Protocol (UDP) floods, Smurf attacks, and buffer overflow.

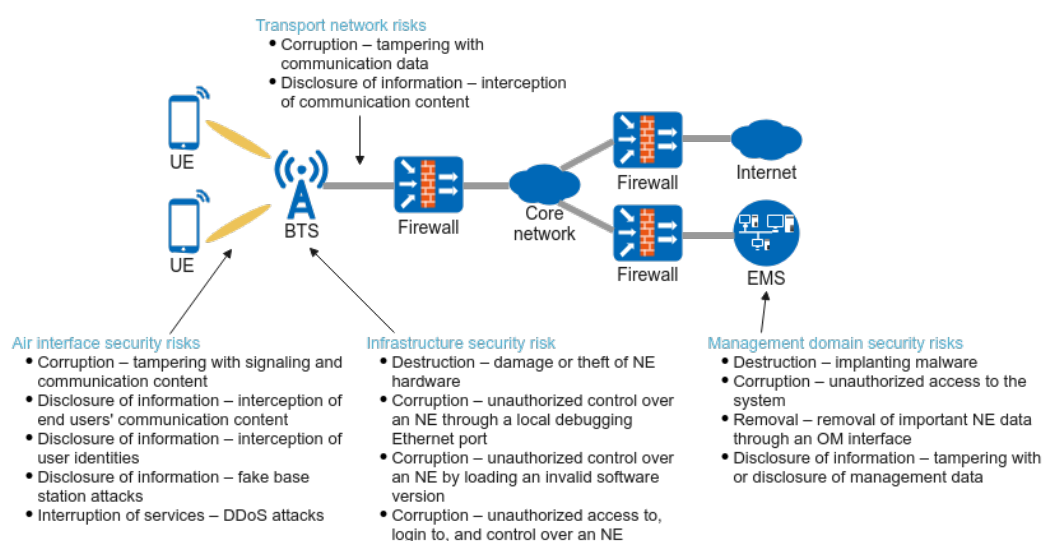
To protect against these attacks, integrated firewalls are usually equipped for NEs, which provide security functions such as packet filtering.

3.1.2 Security Threats on Radio Networks

According to a 3GPP network architecture, a radio network includes UEs, radio access devices (such as base stations), a core network, and an element management system (EMS). In the uplink, the radio access device provides radio access services for the UE and forwards user data of the UE to the core network. In the downlink, the radio access device receives user data sent from the core network, and forwards the user data to the UE over the radio interface.

Figure 3-2 uses base stations as an example to describe the major security threats to radio networks and the corresponding common security threats.

Figure 3-2 Security threats to radio networks



For details about the security solutions, see [4.2 SingleRAN Security Protection Deployment](#).

3.2 Security Dimensions

ITU-T X.805 defines eight security dimensions, which outline security protection measures to be implemented in system security design and deployment. These security dimensions are as follows:

- **Access control**
Access control ensures that only authorized personnel or network equipment can access NEs, stored information, information flows, services, and applications. The examples include the access control list (ACL) function for network equipment, and executable man-machine language (MML) commands allocated to users with different levels of rights.
- **Authentication**
Authentication verifies the identities of the communicating entities and protects against anonymous communications or unauthorized replay of previous communications contents. The examples include password protection for operation personnel and mutual authentication between NEs using digital certificates.
- **Non-repudiation**
The non-repudiation security dimension provides means for preventing an individual or entity from denying having performed a particular action by making available proof of various network-related actions, such as system logs and security audit logs of NEs as well as connection and information exchange between NEs based on digital certificates. It ensures the availability of evidence to prove that some kind of event or action has taken place.
- **Data confidentiality**
Data confidentiality prevents data from unauthorized disclosure and prevents data contents from being understood by unauthorized entities. The examples include SSL encryption of OM data and encryption of IP data between NEs using IPsec.
- **Communications security**
Communications security ensures that data is transmitted only between authorized end points.
- **Data integrity**
Data integrity ensures the correctness or accuracy of data. This protects data from unauthorized modification, removal, or creation and reveals such unauthorized activities. The examples include integrity protection for data transmitted over the radio interface and integrity protection for all IP packets using IPsec.
- **Availability**
Availability ensures that access to network resources, data, hardware and applications is not denied for authorized users even under cyber attacks. An example is that a firewall still provides normal services for authorized users even in the case of DoS attacks.
- **Privacy**

Privacy prevents unauthorized personnel or equipment from deriving user behavior or attribute information by network monitoring. Such information includes the list of websites visited and location information of users. An example for protection is that saving user information in the system is prohibited.

3.3 Huawei Security Principles and Policies

Based on its extensive experience in the development of communications equipment and state-of-the-art industry practices, Huawei has formulated the following security principles and policies:

- **Appropriate security**
Based on the analysis of specific security threats, security design aims to select the most appropriate security measures under a comprehensive evaluation of performance, costs, and commercial effect.
- **Least privilege**
Only the minimum rights and resources required for a task are allocated to users, OM personnel, NEs, programs, and procedures in order to eliminate potential security threats.
- **Collaborative protection**
Communications network security is achieved through collaboration of equipment, management, and OM personnel.
- **In-depth protection**
Huawei provides multi-layer security protection for a SingleRAN system. The in-depth protection aims to eliminate individual security vulnerabilities on the network.

3.4 Huawei Security Management

Modern mobile communications networks and equipment are complex, diversified, and updated frequently. In addition, ever-evolving cyber attacks are continuously emerging, posing a great challenge to the secure running of network equipment. Therefore, security management must be implemented for the networks and network equipment. To combat these attacks in a visible and manageable fashion, Huawei implements the following types of security management on a SingleRAN system:

- **Security policy management**
Provides customized security policies for security domains and equipment. Security policies can be configured and delivered on the unified security management platform.
- **Security event management**
Collects, analyzes, and processes security events that occur on a network, involving collection, filtering, correlation analysis, and browsing of security logs and preventive maintenance inspection (PMI) of equipment. The system takes proper measures to mitigate damage upon detection of any invasions.
- **Security threat and vulnerability management**

Provides vulnerability detection, threat analysis, risk assessment, risk mitigation suggestions, and security reports. Huawei introduces a security management system to effectively handle security threats and vulnerabilities.

4 Security Architecture

4.1 System Security Protection

Huawei's SingleRAN system provides diverse security features to protect against any potential security threats in order to achieve comprehensive system security. [Table 4-1](#) describes the security threats and the corresponding security measures.

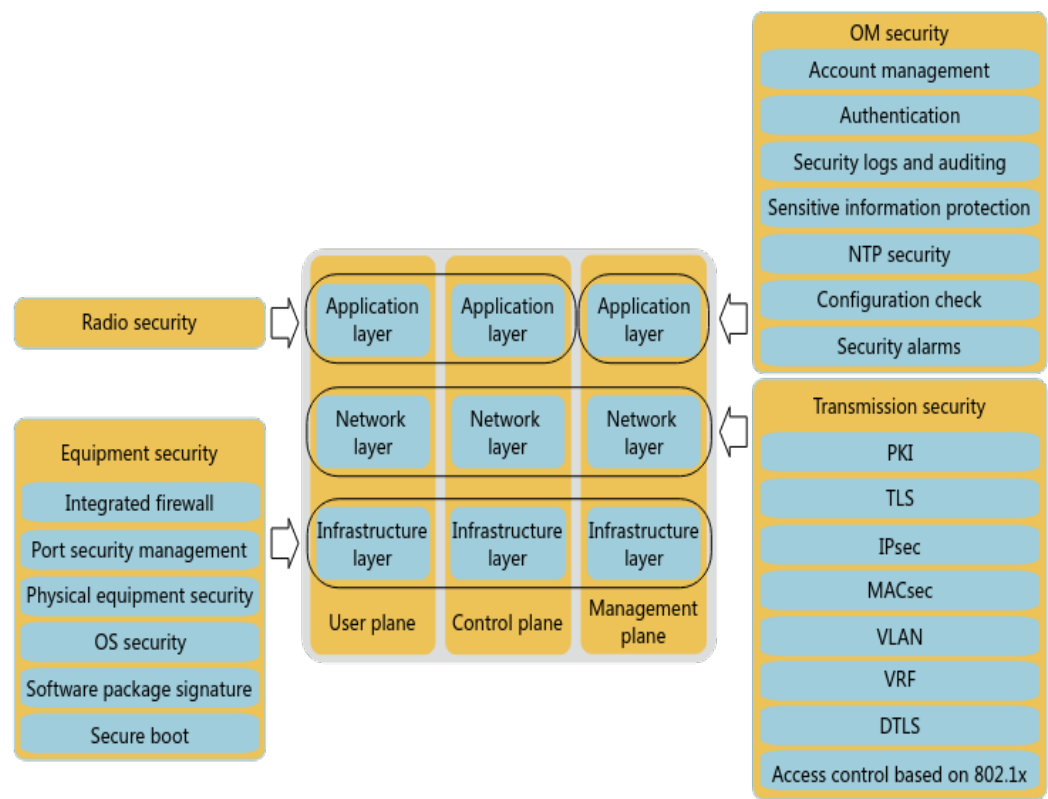
Table 4-1 Mapping between security threats and security measures

Security Threat	Security Threat Description	Security Dimension	Security Measure
Destruction	Damage or theft of the hardware of NEs	Access control	Physical equipment security (locking equipment rooms and outdoor base stations)
	Penetration attacks on any two planes through the other plane among the user, control, and management planes, to destroy or control the system or steal system information	Availability	Integrated firewall and NE emergency mode
Corruption	Unauthorized control over an NE through a local debugging Ethernet port	Access control	Port security management
	Unauthorized control over an NE by loading an invalid software version	Data integrity	Digital signatures of software

Security Threat	Security Threat Description	Security Dimension	Security Measure
	Attacks on an NE using invalid clock sources	Data integrity	NTP security authentication
	Unauthorized access to, log in to, and control of an NE	Access control	Authentication, public key infrastructure (PKI), access control based on 802.1x, and security alarms
Removal	Removal of important NE data, such as configuration files or version information through an OM interface	Non-repudiation	Account management and security logs
Disclosure of information	Unauthorized access to keys used by NEs	Data confidentiality	Sensitive information protection
	Unauthorized access to important NE data transmitted through an OM interface	Communications security	SSL/TLS
	Access to important information by exploiting operating system (OS) vulnerabilities	Availability	OS security
Disclosure of information	Unauthorized access to user communications data through an interconnection interface	Communications security	MACsec
Disclosure of information	Unauthorized access to user communications data through a service interface	Communications security	IPsec
Interruption of services	Service interruption, NE unavailability, or NE performance deterioration by launching DoS attacks or causing OS malfunctions	Availability	Integrated firewall, OS security, security alarm, and NE emergency mode

Security features in the Huawei SingleRAN security architecture are classified into OM security, equipment security, transmission security, and radio security. These security features cover three layers and three planes of the system. [Figure 4-1](#) illustrates the architecture of these security features.

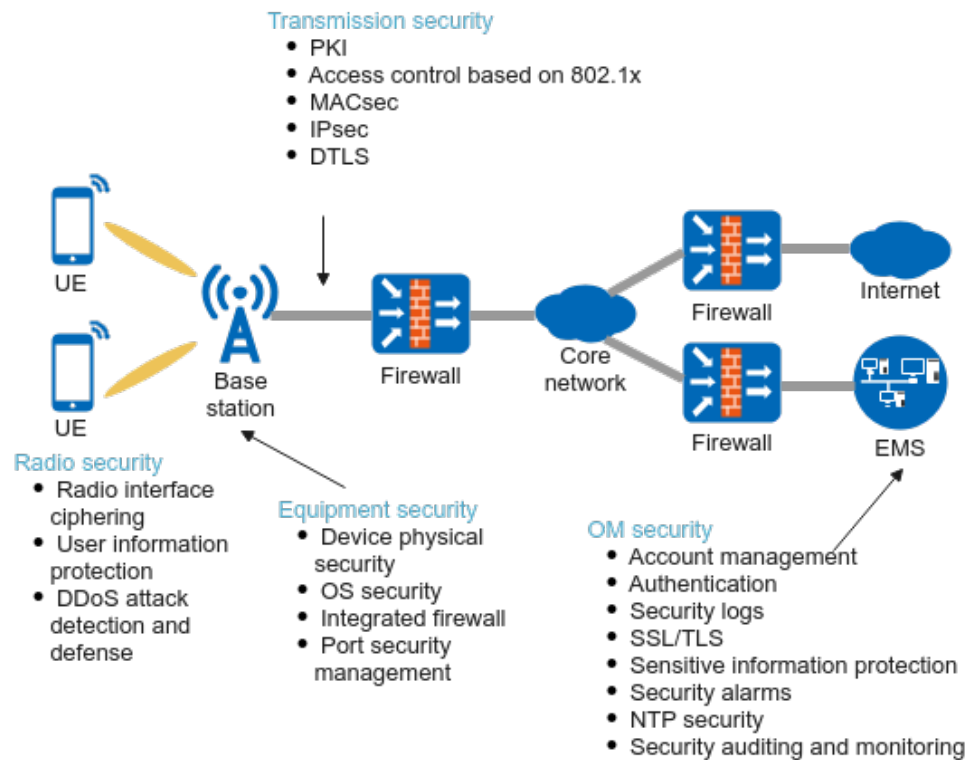
Figure 4-1 Architecture of Huawei SingleRAN security features



4.2 SingleRAN Security Protection Deployment

To address the major security threats on radio networks, Huawei has developed security-related protection functions for radio access devices to protect radio networks. [Figure 4-2](#) illustrates the security protection capabilities of Huawei base stations.

Figure 4-2 Security protection capabilities of Huawei base stations



5 Security Features

The security features described in the following sections apply to the following NEs:

- GSM and UMTS: base stations and base station controllers
- LTE: base stations
- NB-IoT: base stations
- NR: base stations

5.1 Equipment Security

[Table 5-1](#) lists the equipment security measures supported by Huawei SingleRAN.

Table 5-1 Equipment security measures

Equipment Security Measure	Description
Integrated firewall	IP data flows (including OM and service data) to a base station or base station controller are preprocessed by a firewall integrated in this NE. The preprocessing filters out invalid or attack packets to ensure radio equipment security.
Port security management	Huawei SingleRAN equipment provides a communication matrix, which defines open ports complying with protocols, and allows for disabling of unused service ports. For details about the communication matrix, see <i>BSC6900 Communication Matrix</i> in BSC6900 GU product documentation, <i>BSC6910 Communication Matrix</i> in BSC6910 GU product documentation, and <i>3900 & 5900 Series Base Station Communication Matrix</i> in <i>3900 & 5900 Series Base Station Product Documentation</i> .

Equipment Security Measure	Description
Physical equipment security	Equipment must always be securely located in an operating environment that meets the requirements for smooth running. The environment has stable voltage, temperature, and humidity and meets fire protection and other requirements. Huawei SingleRAN equipment supports monitoring of the environment conditions. If any condition becomes abnormal, the equipment generates an alarm and reports the alarm to the operation and maintenance center (OMC) for prompt handling by OM personnel.
OS security	The OS of Huawei SingleRAN equipment uses a mainstream vulnerability scanning tool to prevent high-risk vulnerabilities. In addition, measures such as OS hardening and OS patch installation are also available for the enhancement of OS security.

For details about equipment security, see *Equipment Security*.

5.2 Transmission Security

During data transmission on a network, hackers may intercept key information and use it to launch attacks on NEs. [Table 5-2](#) lists the transmission security measures supported by Huawei SingleRAN.

Table 5-2 Transmission security measures

Transmission Security Measure	Description
Internet protocol security (IPsec)	Huawei base station controllers and base stations support IPsec networking to ensure data confidentiality and integrity and provides authentication protection during data transmission. For details, see <i>IPsec</i> .
Public key infrastructure (PKI)	PKI binds a certificate holder's identity with a public key using signed digital certificates to facilitate the acquisition, access to, and revocation of certificates. Additionally, PKI uses digital certificates and related services to authenticate the identities of communicating entities, which ensures confidentiality, integrity, and non-repudiation of communications data. Huawei base stations support initial certificates and comply with Certificate Management Protocol v2 (CMPv2). For details, see <i>PKI</i> .

Transmission Security Measure	Description
Access control based on 802.1x	Access control based on 802.1x is a prerequisite for Huawei base stations to be connected to a customer network to transmit service data over the ports. This prevents unauthorized equipment from being connected to the customer network. For details, see <i>Access Control based on 802.1x</i> for SingleRAN.
Secure socket layer (SSL)/Transport layer security (TLS)	SSL is a protocol that provides end-to-end communications security between the Transmission Control Protocol (TCP) layer and the application layer. SSL ensures security of application-specific protocols such as HTTP, FTP, and Telnet. For details, see <i>SSL</i> for SingleRAN.
Media Access Control security (MACsec)	MACsec is used to protect data transmitted between UMPT boards interconnected through the Ethernet. For details, see <i>MACsec</i> for SingleRAN.
DTLS(NR)	DTLS is introduced to ensure the security of data transmission based on unreliable protocols, such as UDP. It is recommended that DTLS be used for signaling transmission over the NG-C and Xn-C interfaces to ensure security. For details, see <i>DTLS</i> for SingleRAN.

5.3 OM Security

Table 5-3 lists the OM security measures provided by Huawei SingleRAN.

Table 5-3 OM security measures

OM Security Measure	Description
Account management	Huawei SingleRAN equipment supports both centralized and local account management. - Centralized account management uses operations support system (OSS) software to centrally manage NE users. This ensures a centralized OM policy in the OMC. - Local account management uses the local maintenance terminal (LMT) of NEs to manage local accounts for local maintenance.

OM Security Measure	Description
Authentication	Authentication confirms the identities of communicating entities to ensure communications only between authenticated communicating entities. Huawei SingleRAN supports both the centralized domain user authentication mode and the local user authentication mode. • In centralized domain user authentication mode, only a domain user authorized by the MAE is allowed to log in to an NE. • In local user authentication mode, a local user is allowed to directly log in to an NE using a valid user name and password. Base stations and base station controllers classify accounts into four levels by default based on maintenance operations and their significance. User accounts of different levels are allocated different rights. The systems also support user-defined account levels to facilitate management.
Security logs	Huawei multimode base stations and base station controllers support security logs. A security log records security-related operations and events during normal running or routine maintenance of equipment. The recorded information cannot be modified. Therefore, these logs are used for routine security auditing of a network and help trace security events and issues that previously occurred on live networks and locate their root causes. This facilitates network performance improvement.
SSL/TLS	In SingleRAN, SSL- or TLS-encrypted OM channels between NEs and the OSS and between NEs and the LMT help prevent the disclosure of sensitive information.
Sensitive information protection	In system maintenance and commissioning, sensitive information is processed to protect user privacy. On GSM and UMTS networks, sensitive information about user privacy is pseudonymized. For details, see <i>User Data Pseudonymization</i> in GBSS feature documentation or RAN feature documentation. Sensitive information about user privacy is anonymized on the NR network and is anonymized or pseudonymized on the LTE network.
Security alarms	Huawei SingleRAN equipment supports real-time reporting of security alarms to OM personnel to inform them of security risks and threats. It also provides security suggestions to eliminate security threats on the network.
NTP security	Integrity check and authentication for NTP packets ensure that the packets are not tampered with or forged. This prevents synchronization malfunctions on networks under attacks. For details, see <i>OM Security</i>.

OM Security Measure	Description
Security status auditing and monitoring	During PMI of equipment, the NE information collector (NIC) is used to collect NE data and a special-purpose tool is used to analyze the security status and configurations of live-network equipment at the time of data collection. This helps OM personnel rapidly identify security threats and fix the corresponding vulnerabilities to ensure proper running of the equipment.
Digital signatures of software	Digital signatures protect software against malicious modifications or viruses during software upgrades.
Security configuration check	The security configuration check function checks security-related parameter configurations of an NE according to specified check rules and provides risk descriptions, handling suggestions, and rectification impacts in the check results. OM personnel can then query the check results, perform comprehensive evaluation, determine the parameter reconfiguration policy, and perform reconfiguration to mitigate the device security risks.

For details about OM security, see *OM Security*.

5.4 Radio Security

When a UE accesses the radio network, certain measures must be taken to prevent hackers from intercepting radio signals from the radio interface and cracking communications data. The radio security solution mainly aims at radio interface security.

Radio interface encryption and integrity protection algorithms are used to protect radio transmission between UEs and NEs. For details about radio security, see the following documents:

- *Ciphering* in GBSS feature documentation
- *Data Integrity Protection and Encryption* in RAN feature documentation
- *Radio Security* in eRAN feature documentation
- *Radio Security* in 5G RAN feature documentation

5.5 Specific Security Policies

For details about the default configuration, recommended configuration, and network impact of security policies, see *Specific Security Policies*.

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- ITU-T X.800: Security architecture for Open Systems Interconnection for CCITT applications
- ITU-T X.805: Security architecture for systems providing end-to-end communications
- *SRAN Networking and Evolution Overview*
- *OM Security*
- *Equipment Security*
- *User Data Pseudonymization* in GBSS feature documentation or RAN feature documentation
- *Access Control based on 802.1x*
- *IPsec*
- *PKI*
- *SSL*
- *MACsec*
- *DTLS*
- *Specific Security Policies*
- *Ciphering* in GBSS feature documentation
- *Data Integrity Protection and Encryption* in RAN feature documentation
- *Radio Security* in eRAN feature documentation
- *Radio Security* in 5G RAN feature documentation
- *BSC6900 Communication Matrix* in BSC6900 GU product documentation
- *BSC6910 Communication Matrix* in BSC6910 GU product documentation
- *3900 & 5900 Series Base Station Communication Matrix* in *3900 & 5900 Series Base Station Product Documentation*